



"Tinkering is Underrated": Librarians' Use of Single-Board Computers and Microcontrollers Outside of Makerspaces

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“Tinkering is Underrated”: Librarians’ Use of Single-Board Computers and Microcontrollers Outside of Makerspaces

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Abstract

Purpose – This research seeks to better understand the potential uses of maker-technologies, such as single-board computers and microcontrollers, more broadly within libraries and not simply confined to the makerspace. Through interviews with librarians creating such projects, this study illustrates their successes, challenges, means of acquiring the necessary skills and knowledge, as well as their perceptions of the broader benefits and challenges to other library and information science practitioners.

Design/methodology/approach – This research study employed semi-structured interviews with twelve librarians who have created projects with maker-technologies for broader library use. Inductive qualitative analysis of the interview transcripts was conducted to identify themes of interest to the stated research questions.

Findings – Librarians’ projects included: displaying digital signage, hosting OPAC stations, tallying reference desk interactions, counting patrons at the gate, and monitoring 3-D printing statistics, among others. Participants appreciated the low-cost, flexible, and creative nature of such technologies, and though they also encountered technical and organizational challenges in their use, relayed a potential series of benefits to librarians and library staff were these technologies to be more widely used.

Originality – Although significant research efforts have focused on aspects of makerspaces across all types of libraries, little work has formally collected and assessed library practitioners’ work with maker-technologies outside of the makerspace. Participants help detail the potential benefits of having a deeper understanding of this work and the successes it could bring to librarians’ work.

Keywords Makerspaces, Raspberry Pi, Arduino, Makers, Libraries, Information Technology, Librarian Competencies, Librarian Education

Paper type Research paper

1. Introduction

The library makerspace, along with its associated technologies, educational resources, and events, has rapidly become a commonplace offering for numerous libraries, in both academic and public settings. The many, and often challenging, aspects of creating and running a makerspace have been extensively covered in formal and informal library and information science literature, as well as in workshops, conferences, and tutorials. Though a small subset of librarians and library staff are likely working directly with makerspaces currently, this proliferation of makerspaces means that maker supplies and technologies are available to a

greater number of library practitioners. These maker-technologies include popular “DIY” tools, such as the Raspberry Pi – a low-cost single-board computer originally developed for educational purposes by the Raspberry Pi Foundation – and Arduino microcontrollers – another popular education-focused electronics prototyping platform. Both technologies maintain a longstanding popularity with maker-communities, for their flexible application in countless projects, and have become a standard offering in makerspaces of all kinds. Not surprisingly given the interest in such tools within the computing and library communities, maker-projects within libraries – but outside of the makerspace – have also sprung up in recent years which use these tools to support, enhance, or create novel library services.

The existing body of library and information science research around makerspaces, described further in the following section, indicates that makerspace tools and technologies are commonplace in a large portion of libraries of all kinds, that librarians and library staff often desire more skills and knowledge to work with such tools, and that projects with these technologies outside of the makerspace provide the potential to both solve practical problems and enhance the library’s marketing activities. As maker-technology projects designed for wider library usage are still relatively rare in published literature (compared to the vastly more studied makerspace itself), this study seeks to better understand librarians’ motivations in creating such projects, as well as their successes and challenges, with the longer-term goal of making such work more approachable and feasible to the wider library community. Thus, the goal of this work is to further our understanding of the potential for maker-technologies to be used more broadly within the library, through conducting semi-structured interviews with librarians who have worked on similar projects. To that end, this research study pursues the following research questions:

RQ1 – *How and why* have librarians used maker-technologies (such as the Raspberry Pi and Arduino) in projects outside of the makerspace, and what were the *successes* and *challenges* of these projects?

RQ2 - How did the librarians involved in such projects *gain the necessary skills and knowledge* to work with these maker tools?

RQ3 – What do librarians involved in such projects feel are the general *benefits and challenges* of maker-technologies employed in projects outside of the makerspace?

RQ4 - What *skills and knowledge* do these librarians feel would be necessary for others to know in order to implement similar projects in their libraries?

2. Literature Review

A great deal of prior research has explored the emergence of the library makerspace, in concept and in practice, as well as the makerspace’s natural connection to the library environment and mission. Surveys in recent years have found that maker-technology skills are increasingly desired by librarians (e.g., as found in Burke’s survey in 2016) and have also confirmed the vast popularity of the makerspace across libraries of all kinds. A 2015 survey (Altman *et al.*) found

that 64 percent of responding North American research libraries were in a process of planning for, implementing, or running a makerspace.

As makerspaces become less of a novelty within libraries, research has expanded to cover the related educational offerings and the needs of librarians that work in such spaces and are tasked with educating patrons engaged in making activities. A 2016 survey of library makerspace practitioners (Koh and Abbas) emphasized respondents’ desire for increased technology skills, with nearly half of survey participants citing this need, as well as described their current feelings of being unprepared in this area. Prior research has determined there to be a clear lack of formal education in the area of makerspaces and maker-technologies (e.g., Williams and Folkman, 2017; Moorefield-Lang, 2015; Koh and Abbas, 2016; Maceli, 2019). Horton’s 2019 survey of library staff members working in makerspaces or with related technologies explored their preferences and need for continuing education in this area, finding that most learned the necessary skills to work with these technologies through a variety of online means and workshops, with few having taken a formal course in this area, though all were interested in more educational offerings. To support librarians and library staff new to the area, numerous how-to guides and other practical suggestions around starting or running one’s own library makerspace have been published in recent years (such as Kroski, 2017; Willingham, *et al.*, 2017; Kroski, 2020; and Mathuews and Harper, 2020).

During the same time frame that makerspaces began to garner interest in library communities, many of the same technologies found within them were being employed in solving problems or prototyping solutions to library needs. The technologies found in makerspaces - in particular, low-cost single-board computers and microcontrollers popular with tinkerers and educators, such as the Raspberry Pi (Raspberry Pi Foundation, 2021) and Arduino (Arduino.cc, 2021) - have many potential applications both in and outside of the makerspace. Numerous and varied uses of such maker-technologies have been documented across different aspects of library work, in formal and informal publications. Tergla (2013) hosted information kiosks and signs with Raspberry Pis, while Cooper and Knight (2014) used Raspberry Pis as low-cost OPAC terminals. Ribaric (2018) described using maker-technologies to increase automation of reference statistics collection, while Pham *et al.* (2018) used Raspberry Pis along with cameras and a wheeled robot to automate the monitoring of book locations in their large library system. Kung (2018) prototyped a system with Raspberry Pi and Arduino that could be used to collect and visualize noise levels in an academic library, with the goal of creating a better environment for patrons.

Creative and practical projects with these technologies have continued to emerge at a steady pace in recent years. Dillalogue and Koehn (2020) describe using a Raspberry Pi and RFID reader to create a “Tap In/Tap Out tool” which allowed their academic library to collect demographic data useful to their assessment efforts. Williamson and Becker (2019) created the poemBot project, a Raspberry Pi based system that printed poetry onto receipts, intended to raise awareness of physical computing and the public poetry initiative at the at the University of Idaho Library. And at the Thomas Moore College library, Wells (2020) described leading the library administration through a digital modernization plan, in which Raspberry Pis were used as hosts for OPAC computers and self-checkout stations, as well as to run digital information displays,

allowing the library to significantly reduce their technology costs. The bulk of these projects employed the Raspberry Pi, as mentioned earlier, which is a low-cost single-board computer originally developed for educational purposes by the Raspberry Pi Foundation. Technical components typically found in a makerspace were well-represented in the projects discussed, including microcontrollers, a variety of sensors, robotics, cameras, and user interface components (such as buttons and screens).

These projects, and no doubt others that have not been formally published, provide a glimpse into the many creative and meaningful projects that might be accomplished in the library space were making-technologies to move into broader library work. Beavers, *et al.* (2019) advocate for expanding making beyond the library makerspace itself, to a broader maker culture, experimenting with placing maker-technology in visible public locations, forming interest groups, hosting pop-up events, as well as other initiatives. As was the goal of many of the maker-projects detailed above, these not only fulfilled a practical (or aspirational) library need, but also piqued patrons and colleagues' curiosity by placing these maker-technologies in more public view.

The trend of moving making projects out of the makerspace has been a popular educational technique oriented towards patrons, and also librarians and library staff themselves. Purpur, *et al.* (2016) employed mobile makerspaces as a means of engaging and educating their library colleagues in makerspace technologies and techniques, to overcome potential feelings of intimidation or disconnect between these offerings and the libraries' mission. Brown, *et al.* (2017) present a case study on a successful collaborative staff learning model, based on principles and the ethos of the maker movement, designed to give librarians re-skilling opportunities with the Raspberry Pi and Arduino. This work, and others, indicates a clear interest in deepening the relationship between librarianship and the work of making.

3. Research Method

The research study design consisted of semi-structured interview sessions with librarians who had worked on maker-technology projects outside of the makerspace context, with the goal of understanding their motivations, successes and challenges, prior education in this area, and thoughts about broader use of these technologies. All sessions were held remotely over Zoom, and included responding to a demographic survey form, as well as a series of interview questions (shown in Appendix A). Participants for this study were required to be 18 years of age or older, currently or previously employed by a library, and to have created one or more projects using physical computing devices for use within their library. Participants were recruited via e-mail solicitations posted to listservs oriented towards information professionals and technology-focused library roles. Participants were selected for inclusion in the study on a first-come, first-serve basis and received a gift card for their participation in the study.

In each individual session, participants were first asked to read and sign a consent form and then complete a demographic survey. The demographic survey collected basic job and educational experiences, as well as asked them to rate their familiarity with a series of common maker-technologies and describe their reasoning for choosing to use these tools. The researcher then

posed the list of questions presented in Appendix A to the participant. All participants’ sessions were audio-recorded, with transcripts generated automatically post-session through Zoom Cloud. Each interview took approximately 45 minutes.

The transcribed recordings of participant interviews were analyzed using inductive qualitative analysis to code the transcripts and identify concepts of interest to the research questions. Sessions were conducted concurrently with data analysis until a point of theoretical saturation was identified by the researcher.

4. Results

A total of twelve (12) librarians participated in the research study, with interview sessions held between April 2021 and June 2021 over Zoom. Participants reported having roles as Library Managers (4 participants), Digital/Web Services Librarians (5 participants), Systems Developers/Analysts (2 participants) and Systems Librarians (1 participant) at the time of working on the maker-related projects of interest to this study. Of the twelve librarian participants in the study, four fell into the 18-40 age range, with the remaining eight in the 41-60 age range. Most participants worked in academic libraries (9 participants), while two participants worked within a university IT services department and one participant worked for a national non-profit organization. Most participants were located in North America, with two participants in Britain. All participants held an MLIS/MSIS/MLS degree, while six participants had a dual master’s degree and two participants held doctoral degrees.

4.1 Initial Survey Results

Participants were first asked how often they used single-board computers or microcontrollers to solve a problem within their library. Most participants responded rarely (7 participants), followed by a few times a month (2 participants), daily (2 participants) and a few times a week (1 participant). The survey then asked participants to rate their familiarity with specific maker-technologies (results in Figure 1, below), as well as to list any additional technologies that had been useful in their work. Participants had the highest hands-on familiarity with single-board computers, microcontrollers, soldering, and programming (through the Arduino IDE and/or popular scripting languages such as Python). The least familiar technologies included: electronics kits, circuit diagramming software, and visual programming interfaces.

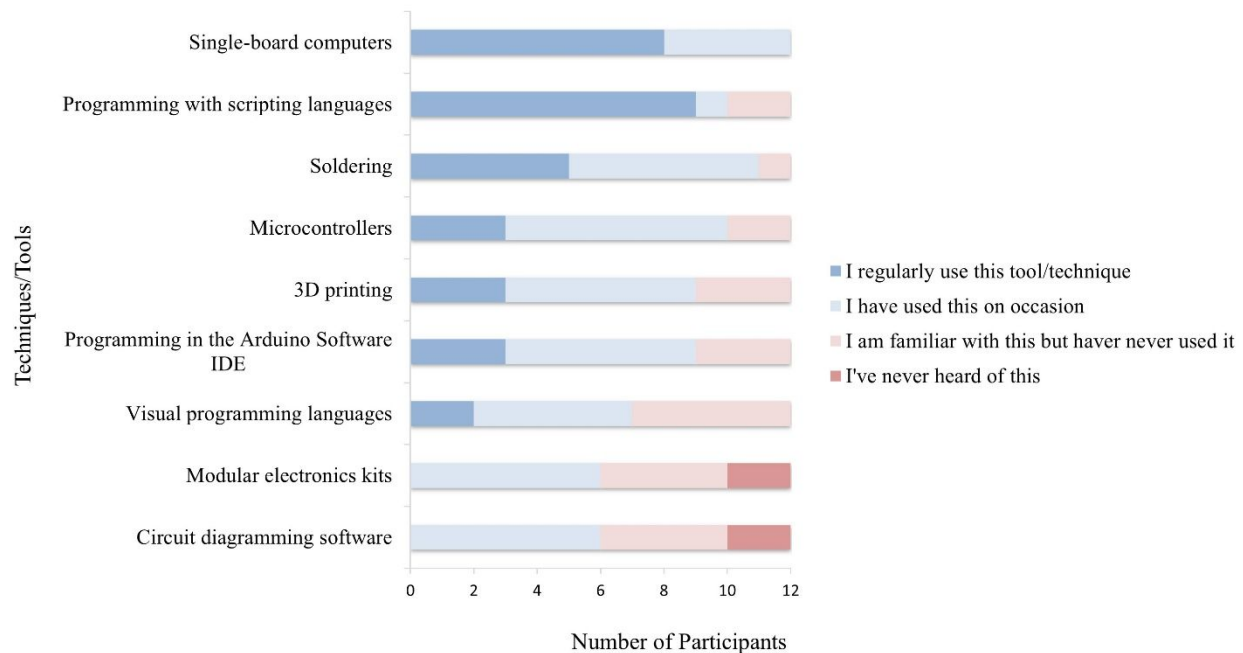


Figure 1: Participant familiarity with popular maker-technologies (N=12), ordered from highest familiarity at the top of the diagram to lowest familiarity at the bottom.

Additional tools/techniques that participants used in their physical computing projects included: laser cutters, humidity/temperature sensors, LEDs, and digital video cameras. Participants also mentioned using specific microcontrollers such as the BBC micro:bit, and LilyPad wearables. Participants related that they chose to use the specific tools/techniques on their projects because of: prior familiarity with the tool/technique (6 participants), low cost (6 participants), availability (3 participants), their own interest in the tool/technique (3 participants), applicability to the project (2 participants), the tool's flexibility (2 participants), ability to support rapid prototyping (2 participants), the importance of using open source software (1 participant), the existence of a strong community of other practitioners (1 participant), and for later use to teach students fundamentals (1 participant).

4.2 Interview Results

In total, there were 23 physical computing projects discussed by the twelve participants. The Raspberry Pi single-board computer was the basis for nearly every project described, except for two projects that employed Arduino micro-controllers. Five of these projects were used to display digital signage to library patrons, including important evergreen information, library events and updates, room reservation status, and available equipment for check-out. Two projects discussed using physical computing devices as web servers to store and facilitate downloading of digital files, while one project was used as a web server for a development testing environment.

Projects also involved tallying reference desk statistics (2 projects), temperature monitoring in the library (2 projects), replacing OPAC terminals (2 projects), gate counting library patrons (2

projects), and monitoring 3-D printing statistics (3 projects). Other projects discussed included a web crawler for archiving the library website, tracking library patron analytics (1 project), monitoring in-use library items (1 project), and a poem printer (1 project). Most participants (8 participants) did not have a formal makerspace in their library at the time of their projects, while four participants did.

Librarians’ motivations to create these projects (Table 1, below) included: the desire to leverage or explore the benefits of maker-technologies (e.g., low cost, easy-to-maintain, etc.), improve existing library processes, improve the patron experience, and replace or supplement existing systems. As detailed in Table 1 (below), the projects improving patron experience and library processes focused on a few areas such as gate counting, collecting reference statistics, improving patron understanding of available items, and increasing the number of catalog search stations. The low cost of maker-technologies was particularly motivating for one participant who described how budget constraints pushed them towards solutions such as the Raspberry Pi:

“We try to find a lot of solutions that use economically accessible and affordable equipment to try to solve a lot of our technological problems. So one of those solutions was using Raspberry Pis because Raspberry Pis are very affordable and it doesn’t take a lot to get them to start a project because there are so many other libraries using Raspberry Pis as well.” [P1]

Several librarians reported an emergent process of determining an appropriate use for the Raspberry Pi, as opposed to starting a project with a fixed goal in mind. A participant related the exploratory nature of their use of the Raspberry Pi, stating that *“our boss also had a sort of ‘play’ budget, so the original Pis were bought just to see what we could do with them so there wasn’t a project in mind.” [P5]* Another participant echoed this approach, saying they were *“interested about these devices and talking about them and just sort of seeing how we can use them in our environment and we kind of worked backwards from there.” [P7]*

<<INSERT TABLE 1>>

Table 1: Librarians’ motivations in creating projects using maker-technologies within their libraries, outside of the makerspace (N=12)

Only a small number of the participants’ projects discussed were still in use at the time of the research study in spring/summer of 2021. Of the projects still in use, two were digital signage projects, one was a 3-D printing project, one was a library patron analytics project, and one was a poem printer. Many projects were described by participants as intentionally transient in nature, and served as a short-term prototype, proof of concept, or an experimental idea. A few projects remained in the conceptual stage, though there was interest expressed in eventually developing them.

Participants described numerous successes and challenges relating to their projects (summarized in Table 2, below). The successes included: improved patron experience, organizational improvements (such as saving on costs), technical successes (such as increased portability and space-saving designs), and multiple personal/professional successes. Most participants spoke at length about the personal and professional successes these projects afforded them, including being able to work on a rewarding and creative project, as well as publishing/presenting their work and expanding their social network to meet others with these interests. Several participants spoke of the rewards of cobbling together a working solution in a low-budget environment:

“...any project that we had that wasn't grant supported was sort of bootstrapped together. You know - find a spare monitor, spare keyboard, spare mouse - so in that kind of environment to just set up this project, just like a pilot project, it felt like a success too” [P4]

“...we basically were looking for ways to address the needs that we thought that the patrons had that we didn't have the money to address with a library budget” [P10]

Several participants related the success of being able to publish and present about their project to the wider LIS community; while some had published supplemental code and instructions, there was a hesitancy in others to do so (e.g., *“I present about it, I write about it, and then I don't share any of the code. I think we're all very self-conscious about messy code that's copy-and-pasted from stack overflow and other places” [P8]*).

<<INSERT TABLE 2>>

Table 2: Librarians' successes and challenges encountered in their maker-technology projects (N=12)

As to the project challenges experienced, these were primarily technical and organizational in nature (Table 2, above). Participants detailed ongoing issues in troubleshooting and maintaining these projects, particularly while being the sole person working on the project, as the main technical challenge encountered in developing and running these projects. As one participant stated:

“Somebody figures that [project] out, and then you need that person to be there. And if you don't have that knowledge transferred to other people in institution, or if you're in a small library where there is only one digitally minded person can be a big problem, if you have all these solutions locked up” [P12]

On the organizational side, lack of support/acceptance from other colleagues and circumstances related to the COVID-19 pandemic were the most frequently mentioned challenges. Explaining the issues around support from colleagues and the library administration, one participant mentioned that *“when there was a choice between a commercially available or a vendor product and something that one of the librarians tinkered and put together there was a general*

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3 *preference for the vendor product.” [P2] Even the largely successful and well-accepted projects*
4 *came with their own challenges in making the solution available for wider use. As one*
5 *participant related “The usual sort of like challenge of a successful project where you suddenly*
6 *realize that it is successful, people are wanting to use it, and then you have to look at it like*
7 *‘Okay, how do we make sure we can scale it up so that people can use it?’” [P6]*
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10 Participants were asked to relate their recollections of how and where they learned the necessary
11 skills to complete the maker-technology projects they described. Participants recalled initially
12 learning about physical computing through having computer/technology hobbies (5 participants),
13 their educational background (3 participants), personal (2 participants) and professional
14 relationships (1 participant), and shared communities (4 participants), including making, gaming,
15 and other online communities. Participants also noted learning about these technologies through
16 magazines and podcasts (2 participants), an interest in incorporating these technologies into their
17 library (3 participants), keeping up with emerging technology for their professional role (2
18 participants) as well as an interest in staying abreast of maker culture trends (1 participant).
19 Given the mid-career age range of many participants in the study, many had been involved with
20 computing for decades and thus remembered the impact of the development of the Raspberry Pi,
21 in particular. As one participant explained:
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25 *“I remember the announcement of the original Raspberry Pi, and it being basically a sort*
26 *of watershed moment of low-cost, low-power computing that it... hit the hobbyist world*
27 *with a sledgehammer, because nothing had ever come quite so close to being a fully*
28 *featured computing platform at a sub \$50 price, it was it was sort of an extraordinary*
29 *thing when it first launched.” [P3]*
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32 Participants reported acquiring the skills and knowledge to work with these technologies largely
33 through hands-on experimentation with the tools (8 participants), followed by learning-by-doing
34 while constructing projects for their libraries (5 participants), shared communities with similar
35 interests (3 participants), their educational background (3 participants), personal relationships (2
36 participants), and online learning resources (1 participant).
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39 Participants were asked what they felt the broader benefits and challenges of working with
40 maker-technologies outside the makerspace would be for other librarians and library staff. All
41 participants cited several benefits and challenges (presented in Table 3, below), with notable
42 agreement around the challenges.
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51 Table 3: Potential benefits and challenges of maker-technologies employed outside of the
52 makerspace by librarians and library staff, as perceived by those having completed such projects
53 (N=12).
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3 The positive potential benefits included increased exposure to programming and the Linux
4 operating system for librarian and library staff, improved flexibility and control of project
5 components, as well as organizational benefits such as lowering the financial risk of prototyping
6 potential solutions. As one participant stated, *"We could try something and see if it worked as an*
7 *idea before we had to commit to that kind of [cost] outlay"* [P9]. Another noted that, *"...when it's*
8 *just a Raspberry Pi or an Arduino or one of the other single-board computers and it's like –*
9 *'yeah, it's a little bit annoying if I break it, but it's not the end of the world'".* [P6]
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12 The chance to collaborate with inter-disciplinary colleagues and maker-communities, as well as
13 expose patrons to maker-technologies were also notable benefits:
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16 *"A benefit is definitely the affordability and the wide community online of other people*
17 *who are using the same technologies and sharing code and tips and tricks. And just like a*
18 *sort of general excitement around the possibilities for how you can use microcomputers,*
19 *circuits, and resistors... So that I see as a draw for students and other patrons, and it*
20 *also sort of shows off the different areas of librarianship that the public maybe doesn't*
21 *know about."* [P4]
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24 Though patrons are generally the focus of maker-related educational events, one participant
25 described the benefits to librarians and library staff themselves in exploring these technologies.
26 While the participant was teaching a workshop to librarian colleagues on Arduino, they noticed
27 one group's success in coding:
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30 *"I just remember sitting down with two librarians at their computer and they were like,*
31 *'so, how would I change the light or the timing'...and I showed them that, and then they*
32 *were able to do it themselves again and they were like jumping for joy. Like they finally*
33 *got that they could do code and it wasn't this mythical brain bending thing, it was*
34 *accessible, and I just really loved that idea."* [P10]
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37 On the challenges side, most were centered around barriers to adoption of these technologies by
38 librarians and library staff. The most common challenges noted by participants were lack of
39 familiarity with and technical knowledge about such technologies, and lack of understanding of
40 use case scenarios for such technologies. As one participant said – *"[Raspberry Pis] weren't a*
41 *single-use obvious device, they took a little more thinking about than people were comfortable*
42 *with"* [P2]. The flexibility of such devices, which was also viewed as a benefit, posed a
43 challenge to use when asking novices in the area to both grasp the technology as well as make
44 the leap as to how they may be used in libraries. Several participants spoke to this issue:
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47 *"...there are so many different projects but you really have to drill down to what are the*
48 *practical applications [in libraries] to these kinds of devices".* [P1]
49

50 *"The challenge...is you need someone that knows how these things work. To be able to*
51 *draw the connections between, you know, a \$50 raspberry pi with some buttons, to this*
52 *challenge we have with keeping track of reference transactions adequately..."* [P7]
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55 Participants also discussed the need to overcome the intimidating perception of maker-
56 technologies as well as encourage librarians to try new technologies, particularly ones which do
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not immediately work and may appear unpolished compared to third-party commercial solutions. One participant also detailed the isolation that those in smaller settings may encounter:

“...for people working in a smaller library who don't have other people working on the same kind of work, it can be challenging to figure out how to do something when you're working with your hands yourself. So it's the kind of technology that really lends itself well to communities and workshops and makerspaces, but not necessarily for tinkering on your own, especially if you're new to the technology.” [P4]

Participants felt a range of skills, knowledge, and attitudes were necessary in librarians and library staff in order to work effectively with these technologies (Table 4, below). The most popular (and non-technical) suggestion was the need for the willingness and drive to independently learn and tinker with such technologies on one’s own. Notably, however, a computer science or technologist background was not one of the necessary elements. One participant stated that:

“...you don't have to have a background in IT, or computer science. I am not a good programmer and, you know, a lot of it is copy-and-paste and you get it to work. And when it does, and when it continues to work, you can do some pretty great things. It's just that willingness to try things out, experiment and tinker. And tinkering is underrated.” [P8]

Participants did mention that a general exposure to scripting languages and coding would be helpful, as well as more comfort with experimenting with computing devices, including accepting potentially “breaking things”, and the persistence to keep trying when ideas failed. Several participants suggested that support for such activities was needed, either through formal LIS programs, mentorship and training programs, or simply a group of like-minded individuals.

<<INSERT TABLE 4>>

Table 4: Skills and knowledge recommended for librarians and library staff working with maker-technologies, as suggested by librarians having completed such projects (N=12).

5. Discussion

The first research question in this study explored - *how and why have librarians used maker-technologies (such as the Raspberry Pi and Arduino) in their libraries outside of the makerspace, and what were the successes and challenges of these projects?* It was clear from the variety of projects that participants recounted (and those detailed in the literature review) that there are numerous possible applications of these maker-technologies in broader library work, particularly for projects based on the Raspberry Pi. These projects included: running display screens, hosting an OPAC station, counting patrons at the gate, tallying reference statistics, creating a web development testing environment, proving an off-grid mobile digital library, capturing temperature and humidity data, among others. While many were motivated by the

needs of patrons or the need to improve librarian and library staff processes, others were more emergent and spontaneous, with librarians brainstorming ways in which solutions like the Raspberry Pi might be experimented with in library work.

An interesting finding was the number of projects that were intentionally created to fill temporary needs or for experimental purposes. Though the lack of long-standing projects may be viewed as a failure of these technologies in the library context, participants very much emphasized that the flexible, low-cost nature of these tools allows them to prototype a solution rapidly, and then move on to the next challenge or use the proof of concept to advocate for a more formal third-party solution. It was clear from participants that technologies like the Raspberry Pi and Arduino fit these needs well and are uniquely suited to these uses.

Relating to the tendency for the project discussed to be temporary in nature, the project successes that were expressed by participants were often personal or professional in nature. For example, the rewarding feeling of finally getting a solution working, or the chance to connect with like-minded colleagues, give a presentation, or publish a paper. Similarly, the challenges encountered were centered around the human-factors as well, particularly the lack of interest and support from colleagues or the administration, though technical factors such as ongoing maintenance posed significant challenges as well.

Next this study investigated - *how did the librarians involved in such projects gain the necessary skills and knowledge to work with these maker tools?* This question was of particular interest given that prior research has noted a general lack of educational support for librarians and library staff working in makerspaces (e.g., Koh and Abbas, 2016), and working with these tools outside of such spaces presumably requires the same skillsets. Formal and continuing educational opportunities played a relatively small role in these librarians' knowledge and skill acquisition in this area. Nearly all participants had been involved with and interested in the field of computing throughout their careers (and were in technology-focused librarian roles) and were thus naturally gaining awareness of these new maker-technologies as they were released, through communities of practice, social media, or other interest groups. Many participants worked in libraries that had tools such as the Raspberry Pi readily available, or had purchased one for personal experimentation, which further lowered the barrier to learning hands-on skills with these systems.

The third research questions addressed - *what do librarians involved in such projects feel are the general benefits and challenges of maker-technologies employed in projects outside of the makerspace?* Clearly, as explored around the first research question, there were numerous opportunities to employ these technologies within libraries, however, no project was immune from challenges encountered along the way. The most daunting challenges that emerged were again focused on the human side, on the need for librarians' acceptance, interest, and willingness to learn new technologies. However, one participant spoke of the deep rewards in overcoming such barriers and learning about these technologies and their potential uses:

"In order for [maker-technologies] to be applied, someone has to have a use case scenario, and so knowing that they exist, and knowing that things are possible with them,

it allows librarians to walk around their library and be less enclosed in the existing solutions to problems... It allows you to see your library with different eyes, it allows you to look at the space and look at the things that you want to happen in that space with a more flexible set of solutions” [P2]

Another participant noted that maker-technologies allow libraries to fully control their own software, hardware, and data:

“In the modern era, the benefit is that if you control the hardware and the software, you also control that data, and if you control the data, that data isn't a risk to your patrons or your organization ...Building what you need rather than purchasing it from a vendor who has secondary motives is a thing that I think should speak to more libraries.” [P3]

This need for control is of ever-increasing importance in an age of data leaks, breaches, and privacy and security threats. This may also suggest a way to increase interest in these tools by highlighting these themes that are of vital concerns to libraries and librarians, as libraries continue to tackle educating patrons on data and privacy literacies.

Lastly, the final research question investigated - *what skills and knowledge do these librarians feel would be necessary for others to know in order to implement similar projects in their libraries?* In response, participants surprisingly discussed relatively few hard, technical skills, advocating instead for a general familiarity with programming, and prioritizing the attitudinal and motivational aspects. It is interesting to contrast these recommendations against the self-professed learning process of the participants themselves, as much of the knowledge/skill acquisition suggested by participants were similar to their own often unstructured, emergent, and responsive learning processes, which leaned heavily on the preponderance of online communities and freely available educational resources.

This finding presents its own challenges to educators in this area, as the need for a tangible, formal educational experience is much easier to remediate than a broader call to increase interest and motivation in technology continuing education. This suggests that future work in this realm must focus on how to raise awareness of the creative potential of such projects, lower the barrier to entry for library practitioners, and create a willingness to spend time on such work, when many other tasks and needs may be higher priority.

6. Conclusion

Through interviews with librarians working with maker-technologies, such as the Raspberry Pi, outside of the makerspace, this study suggests that numerous benefits exist were these technologies to be more widely adopted by librarians and library staff in broader library work. However, significant challenges were made clear as well, including those of an organizational, professional, educational, and technical nature. It is evident that broader community support and educational efforts could help professionals currently working in this realm, as well as encourage those with an interest in learning these new technologies. Future work will explore how ideas for creative and impactful library-focused maker projects could be more widely disseminated and

explained to the LIS community, such that others could benefit from the personally and professionally satisfying maker work described by the librarian participants in this study.

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Appendix A: Semi-Structured Interview Questions

- How did you learn about single-board computer and microcontrollers initially? How did you acquire the skills and knowledge to work with these technologies?
- Describe the project(s) within your library that you created using single-board computers and/or microcontrollers:
 - What was your motivation? When did this take place? How did you acquire the necessary supplies and funding?
 - What succeeded in your project(s)? What could have gone better?

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- How did you share your project with others in your library or organization?
Within the larger LIS community?
 - Did you make code and/or instructions public? Did others adopt your work that
you are aware of?
 - Are your projects still in use? Do you foresee future work in this area?
- Does your library have a makerspace? If applicable, what relationship did your project(s)
have with the library makerspace?
- What do you see as the benefits and challenges of maker-technologies used outside of the
makerspace context in libraries?
- What skills and knowledge do you think librarians and library staff would need to
implement similar projects in their libraries?

Library Hi Tech

Theme	Code (Frequency)
<i>Leverage benefits of maker-technologies</i>	Low maintenance (3)
	Easy setup (3)
	Compact size (2)
	Reliable (1)
	Low cost (1)
	Explore emerging technology (2)
<i>Improve existing library processes</i>	Increase efficiency of tallying reference statistics (2)
	Accessible screen-display solution for staff (2)
	Capture data on library space usage (1)
	Create and control web archiving (1)
	Track in-use books and materials (1)
<i>Improve patron experience</i>	Increase visibility of checked-out/available items (3)
	Provide catalog search stations near stacks (1)
	Provide digital library resources in off-grid areas (1)
<i>Replace or supplement existing systems</i>	Replace gate counter (2)
	Monitor micro-climates within library (2)

Table 1: Librarians' motivations in creating projects using maker-technologies within their libraries, outside of the makerspace (N=12)

Successes (Frequency)	Challenges (Frequency)
● Personal/Professional ○ Allowed for creativity (3) ○ Publishing/presenting results of work to LIS community (3) ○ Opportunity to learn (3) ○ Satisfaction of creating functioning prototype (2) ○ Opportunity to experiment (2) ○ Low pressure project (1) ○ Meeting other LIS community members (1) ● Organizational ○ Cost efficient (2) ○ Avoid purchasing third-party system (1) ○ Ability to create project from spare equipment (1) ● Technical ○ Low maintenance (2) ○ Energy-saving (1) ○ Portability (1) ○ Space-saving (1) ● Improved patron experience ○ Better information dissemination to patrons (1) ○ Decreased visits to reference desk (1)	● Technical ○ Difficulty troubleshooting/maintaining system (5) ○ Limited to on-campus use (1) ○ Feasibility of training other staff/librarians (1) ○ User/human error (1) ○ Scaling up for wider usage (1) ○ Poor aesthetics (1) ○ Ultimately was easier to employ non-technical manual solution (1) ● Organizational ○ Lack of project interest/acceptance from colleagues (3) ○ COVID-19 pandemic disruption (2) ○ Lack of other colleagues to help maintain and troubleshoot (2) ○ Supply procurement (1) ○ Developing necessary policy (1)

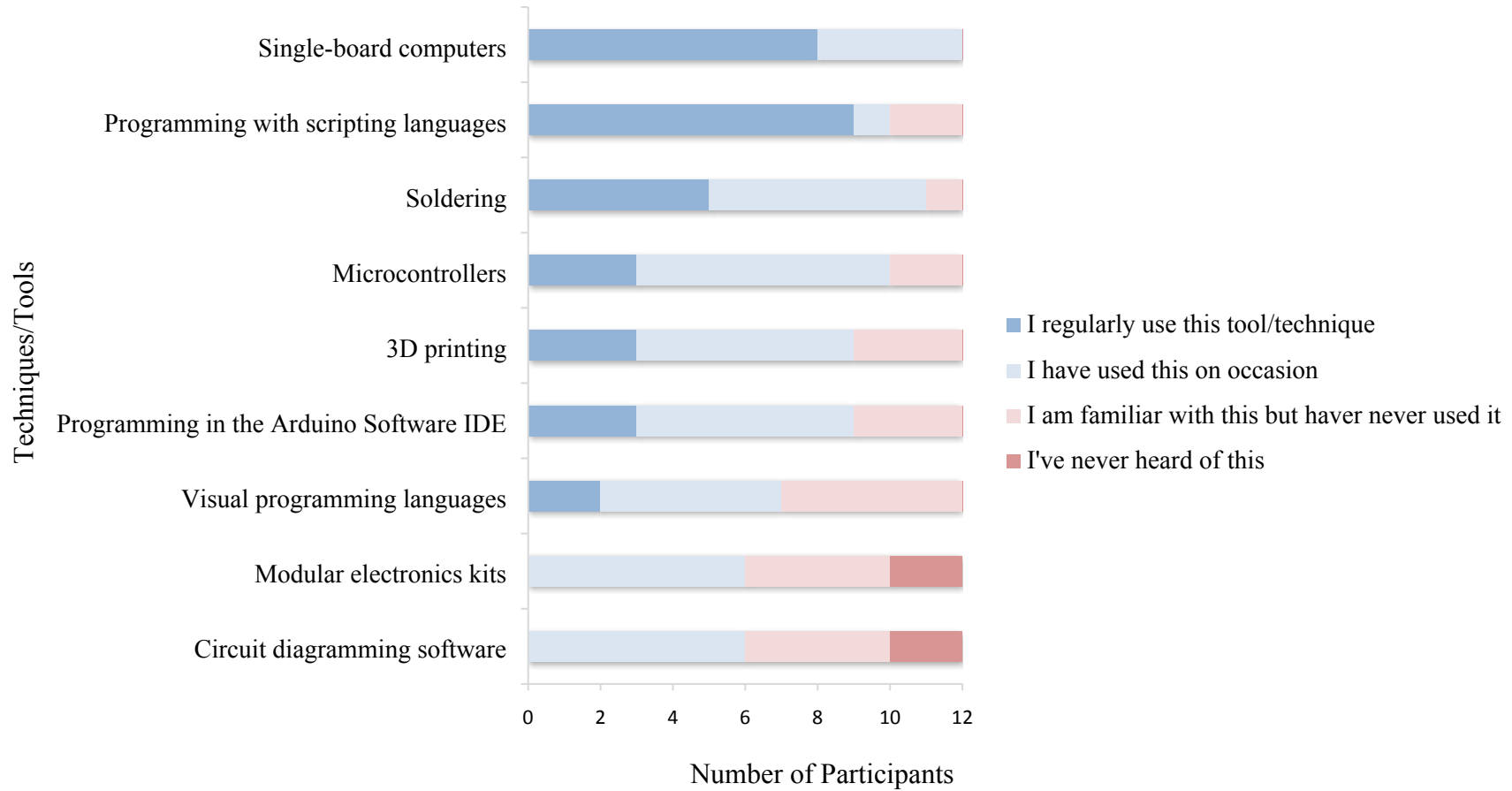
Table 2: Librarians’ successes and challenges encountered in their maker-technology projects (N=12)

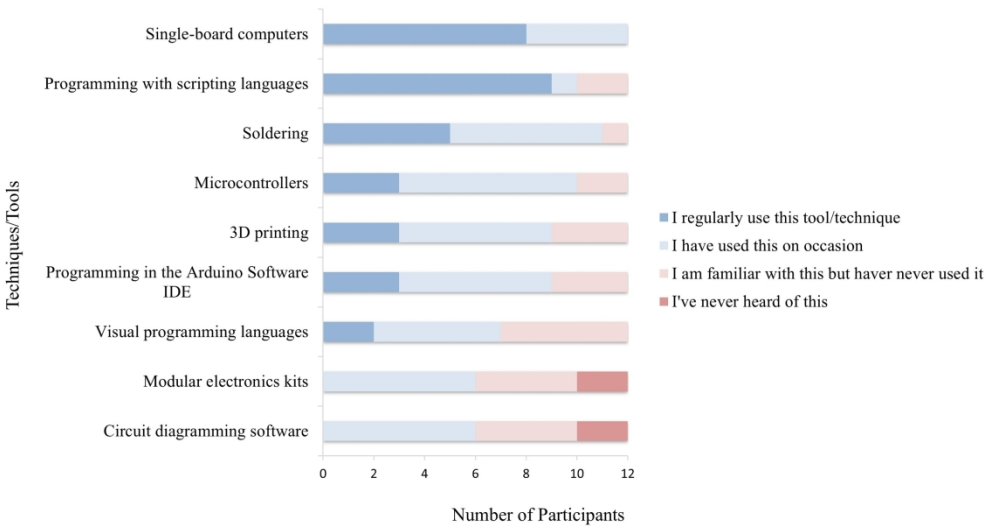
Potential Benefits (Frequency)	Potential Challenges (Frequency)
● Technical ○ Learn Linux operating system and coding (3) ○ More control of software/hardware used in library (2) ○ Ability to rapidly prototype (1) ○ Lower electricity usage (1) ○ Flexibility of use (1) ● Organizational ○ Low financial risk in experimentation (3) ○ Affordability (1) ● Community ○ Facilitates collaboration with interdisciplinary colleagues (2) ○ Learn from large community of practitioners (1) ○ Empower librarians and library staff to create their own projects (1) ● Patron experience ○ Capitalize on excitement around maker-technologies (1) ○ Expose patrons to uses of such technologies (1)	● Librarian/staff adoption ○ Lack of familiarity with and technical knowledge about such technologies (5) ○ Lack of understanding of use case scenarios for such technologies (5) ○ Getting librarians/staff to try new technologies (2) ○ Technologies perceived as intimidating (2) ● Technical ○ Informal look of technologies and enclosures (1) ○ Lack of turn-key solutions (1) ● Organizational ○ Lack of time for necessary experimentation (1)

Table 3: Potential benefits and challenges of maker-technologies employed outside of the makerspace by librarians and library staff, as perceived by those having completed such projects (N=12).

Skills/Knowledge Necessary (Frequency)	
● Technical	
○ Some exposure to scripting languages and coding (4)	
○ Understand broadly what maker-technologies can do (1)	
○ Programming concepts specific to micro-controllers (1)	
○ Exposure to Linux operating system (1)	
○ Experience with system administration (1)	
● Personal/professional	
○ Willingness and drive to independently learn and tinker (9)	
○ Comfortable experimenting (5)	
○ Interest, enthusiasm, and curiosity about technology (4)	
○ Acceptance of “breaking things” and failure (3)	
○ Cooperating with other colleagues and departments (1)	
● Organizational support	
○ LIS programs exposing students to these topics (1)	
○ Mentor or training program (1)	
○ Larger community or group of people to make with (1)	

Table 4: Skills and knowledge recommended for librarians and library staff working with maker-technologies, as suggested by librarians having completed such projects (N=12).





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